

NANYANG TECHNOLOCAL UNIVERSITY
School of Electrical & Electronic Engineering

Guidelines for MSc Dissertations

FORMAT

1. The dissertation should be typewritten in English, in the third person and with one and a half or double spacing between lines. There is no prescribed length to the dissertation; however a good guide would be around 60 to 100 pages printed in font size 12, preferably printed double sided. You are strongly encouraged to use your own words instead of copying directly from books, technical reports, etc. The Standard International Units (SI) should be used.
2. White A-4 size bond paper of at least 80 g weight should be used (applicable for hard copy of the dissertation that requested by the examiner).
3. The margin on the left-hand side must be about 3.5 cm. On the top, bottom and right hand side, a margin of 3 cm is recommended.
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5. The contents of the report should be presented in the following order:
 - (1) Title page
 - (2) Statement of Originality
 - (3) Supervisor Declaration Statement
 - (4) Authorship Attribution Statement
 - (5) Table of contents
 - (6) Abstract
 - (7) Acknowledgment page to give recognition of any advisory or financial assistance received in the course of the work on which the report is based *(optional)*
 - (8) Acronyms *(optional)*
 - (9) Symbols *(optional)*
 - (10) List of figures
 - (11) List of tables
 - (12) Introductory chapter
 - (13) Text chapters
 - (14) References
 - (15) Appendix/appendices *(optional)*
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8. Samples of the cover page, title page, 3 declaration statements and Dissertation format, as well as an extract on the method of listing references, are attached in Appendices 1 to 6 respectively.

STANDARD OF WRITING

The dissertation should be logically laid out. The writing should be in grammatically correct, clear and concise English. There are many books in the NTU library, providing guidance on writing technical reports.

Any attempt to provide guidance or feature common errors here will run into many pages. So we will refrain from doing that except to say one thing: if a word (or phrase or sentence) can be deleted without changing the meaning of what you want to say, then delete it.

All figures and tables should be numbered sequentially, chapter by chapter, and be given a caption. Each must be referred to in the main text, and wherever possible appear near to where it is referred to.

CONTENTS

First and foremost, the contents of your work must be relevant to the MSc Program you are pursuing. Rather than the actual contents, which will vary from dissertation to dissertation and from program to program, this section describes the general areas, which a dissertation should address. While these areas are not necessarily universal, i.e., common to all dissertations, they are more the norm than the exception. The sample Contents page in Appendix 6 gives a pretty good summary on these areas.

1. The **Abstract** should be a short and concise passage on the important work and contributions of the project: the motivation and the problem pursued, the method you employed and the results obtained, highlighting the significant achievements. It should not contain peripheral things like summary of literature review, and it is not good enough to say that a certain issue has been studied without stating the results of the study. Generally, one page is about the right length for the Abstract.
2. The first chapter of the dissertation is almost invariably the **Introduction**. Generally, its purpose is to lead the readers into the problem you intend to attack in the project, to set the scene. The main points here consist of the *background* to the problem and your *motivation* in solving it. This then leads into the *objectives* and the *scope* of the project. It is good to conclude your Introduction with a section on the *layout* of the dissertation. It prepares the readers for what is to come.
3. Then comes the main part of your work. To lay the ground, there should first be a chapter on what has been done before on the problem - a **Literature Review**. This is an important section because it shows that you do not narrowly focus only on what you do, but are aware of the related work elsewhere, some of which might be instructive to your solving the problem. It can also explain why you are taking the direction you do.
4. The next few chapters should describe the work you have done in tackling the problem. There might be a chapter on the fundamental theories relevant to the solution you are pursuing, or the supporting technologies you need in implementing the solution. Then there should be a chapter on the solution itself, followed by a chapter on the results and analysis of the results.
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6. Generally, there should be no more than six or seven chapters in your dissertation. If you have more than that, you should take a close look at its organisation and see if certain chapters can be merged.

SUPPLEMENTARY ELEMENTS

1. Documents that are prepared with the help of other sources should have a list of sources cited. A list of **References** contains only sources the writer quotes directly, takes original ideas from, and refers to in the dissertation should be included. In reports where the subject is primarily scientific, the list of references is the most widely accepted way to cite specific sources.
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Each entry (sample forms, detailed data for references, tables, pictures, questionnaires, charts, maps, graphic representations) in the appendix requires an identifying title. Every entry in the appendix must be referred to in the body of the report. Each appendix must be lettered, beginning with Appendix A. The list of appendices should be appearing in the table of contents following the list of references entry.

REFERENCES

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- [2] Blicq R. S., Technically-Write, Prentice-Hall, 1992.
- [3] Turk C. and Kirkman J., Effective Writing, E & FN Spon, 1989.
- [4] IEEE guideline

Appendix 1: COVER PAGE



DISSERTATION TITLE

AUTHOR'S NAME

SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING

202X

(Submission year)

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A DISSERTATION SUBMITTED IN PARTIAL FULFILMENT OF
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Appendix 5

Authorship Attribution Statement

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Chapter 4 is published as D.T. Murphy, S. Schmid, J.R. Hester, P.E.R. Blanchard, and W. Miiller. Coordination site disorder in spinel-type LiMnTiO₄. *Inorganic Chemistry* **54**, 4636-4643 (2015). DOI: 10.1021/ic502747p.

The contributions of the co-authors are as follows:

- A/Prof Schmid provided the initial project direction and edited the manuscript drafts.
- I prepared the manuscript drafts. The manuscript was revised by Dr Hester and Dr. Blanchard.
- I co-designed the study with A/Prof Siegbert Schmid and performed all the laboratory work at the School of Materials Science and Engineering and the Singapore Synchrotron Light Source. I also analyzed the data.
- All microscopy, including sample preparation, was conducted by me in the Facility for Analysis, Characterization, Testing and Simulation.
- Dr James Hester assisted in the collection of the neutron powder diffraction data.
- Dr Peter Blanchard assisted in the interpretation of the X-ray absorption spectroscopy data and carried out the spectral interpretation.
- Dr Wojciech Miiller assisted in the collection and provide guidance in the interpretation of the magnetic measurement data.

Chapter 5 is published as H. V Doan, B. Yao, Y. Fang, A. Sartbaeva, U. Hintermair, V. P Ting, Controlled Formation of Hierarchical Metal-Organic Frameworks using CO₂ Expanded Solvent Systems. In press, *ACS Sustainable Chemistry & Engineering* (2017). DOI: 10.1021/acssuschemeng.7b01429.

The contributions of the co-authors are as follows:

- Prof Ting suggested the materials area and edited the manuscript drafts.
- I wrote the drafts of the manuscript. The manuscript was revised together with Dr. Sartbaeva and Dr. Yao.
- I performed all the materials synthesis, collected X-ray diffraction patterns and visible light spectra, carried transmission electron microscopy, and conducted data evaluation.
- Dr. Y. Fang conducted the Rietveld analysis of the powder X-ray diffraction data and single crystal structure determinations.
- Dr U. Hintermair conducted the molecular dynamics simulations.
- Ms. A. Sartbaeva prepared the samples for electron microscopy.

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Abstract

Multihop cellular networks (MCNs) incorporate wireless ad hoc networking into *traditional* single-hop cellular networks (SCNs) and thus they enjoy the flexibility of ad hoc networks, while preserving the benefit of using infrastructure of SCNs. In this Thesis, we study the resource allocation problems in MCNs.

Xxxx ...

Acknowledgements (optional)

First of all, I would like to express my sincere thanks and great gratitude to my
parents. ...

Xxx Xxx

November 2010

Acronyms (optional)

2G	Second Generation
3G	Third Generation
ACA	Adaptive Channel Assignment
AP	Access Point
ARS	Ad-hoc Relaying Station
ASP	Adaptive Switching Point
ATDMA	Advanced Time Division Multiple Access
BS	Base Station
CAMA	Cellular Aided Mobile Ad-hoc Network
CBM	Cellular Based Multihop Systems
CDD	Code-Division Duplexing
D-PRMA	Distributed PRMA
DA	Demand Assignment
DCA	Dynamic Channel Assignment

Symbols (optional)

B	channel bandwidth in Hz
C	channel capacity in bps; number of collisions in time slot t
d	distance
D	minimum reuse distance
D_a	average message access delay
D_{id}	inter-datagram-arrival time
D_{max}	maximum tolerable delay for voice packets
D_{pc}	reading time between two consecutive packet call requests

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Chapter 1

Introduction

This chapter

1.1 Motivations

This thesis deals with the problem of the blind multiuser detection for DS-CDMA ...

1.2 Objectives and Scope

The communication channel considered in this thesis is assumed to be slow time-varying,

...

1.3 Organisations

....

Chapter 2

Literature Review

2.1 xxx

2.2 xxx

Chapter 3

SAMPLE

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Chapter 6

Conclusions and Future Work

6.1 Conclusions

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6.2 Recommendation in Future Work

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Appendix (optional)